

AI FAQ ASSISTANT API

4. Project Planning Phase

4.1 Project Activities

The project can be divided into the following activities:

1. Problem identification
2. Requirement collection
3. Technology selection
4. System design
5. Database design
6. API development
7. AI integration
8. Testing
9. Documentation
10. Demonstration

4.2 Team Responsibilities

Possible responsibilities:

Team Member 1

- Requirement analysis
- Documentation

Team Member 2

- Backend/API development

Team Member 3

- Database development

Team Member 4

- AI integration
- Testing

Responsibilities can be changed according to the actual team

4.3 Risk Management

Risk	Solution
API failure	Implement error handling
Database failure	Backup database
Incorrect answer	Improve FAQ/AI data
Invalid input	Input validation
Slow response	Optimize API/database
Security issue	Authentication and validation

- More advanced AI models.
- Integration with websites and mobile applications.
- Automatic FAQ generation.
- Human-agent escalation.

7.9 Conclusion

The **AI FAQ Assistant API** provides an automated approach to answering frequently asked questions. By combining an API backend with AI/NLP processing and a structured FAQ knowledge base, the system can provide quick and consistent responses. The project can be extended to different domains such as education, healthcare, customer service, banking, and business applications.

AI FAQ ASSISTANT API

8. Project Demonstration Phase

During the final demonstration, the project can be presented in the following order.

Step 1 – Introduce the Project

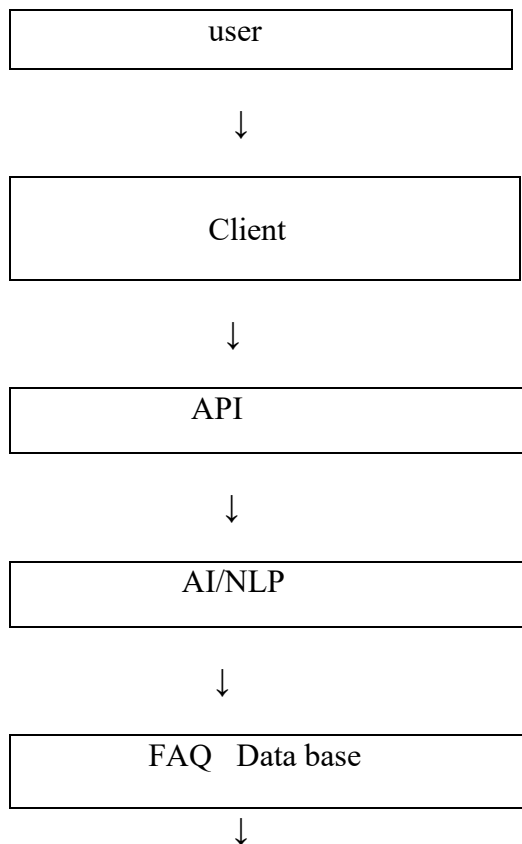
Project Title: AI FAQ Assistant API

Explain:

- What the project does.
- Why it is needed.
- Who can use it.

Step 2 – Explain Architecture

Show:



Answer

Step 3 – Start the Backend

Run the backend application.

Example:

```
Server started successfully  
Running on localhost
```

Step 4 – Open Postman

Create:

POST /api/faq/ask

Enter:

```
{  
  "question": "How can I reset my password?"  
}
```

Step 5 – Send Request

Click **Send**.

Expected result:

```
{  
  "success": true,  
  "answer": "You can reset your password using the Forgot Password  
option."  
}
```

Step 6 – Demonstrate Error Handling

Send:

```
{  
  "question": ""  
}
```

Show:

```
{  
  "success": false,  
  "message": "Question cannot be empty."  
}
```

Step 7 – Demonstrate Multiple Questions

Try:

What are the working hours?

How can I contact support?

How do I reset my password?

Show that the API provides the appropriate responses.

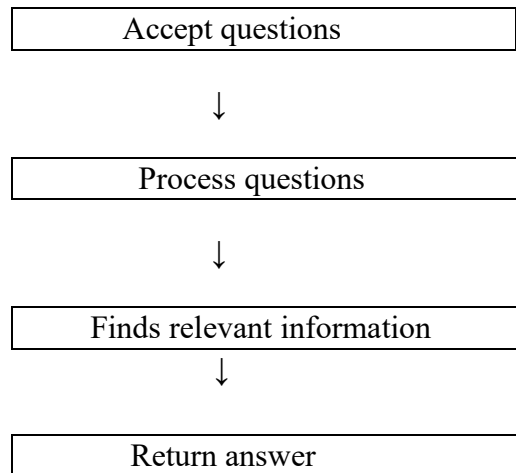
Step 8 – Explain GitHub

Show:

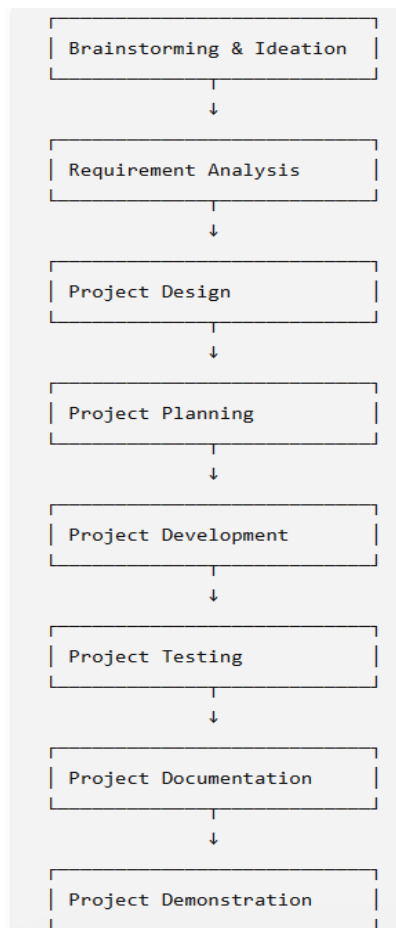
- Source code
- README
- Documentation
- Database files/scripts
- Testing files
- Screenshots

Step 9 – Final Result

Conclude the demonstration by showing that the system:



Final Project Flow



Conclusion

The AI FAQ Assistant API is an intelligent and efficient system developed to automate the process of answering frequently asked questions. The system allows users to submit questions through an API and receive relevant answers quickly. By using AI/NLP techniques and a structured FAQ knowledge base, the application can understand user queries and provide suitable responses.

The project helps reduce repetitive work for support staff, saves time, improves accessibility to information, and provides consistent responses to users. The API-based architecture also makes the system flexible and easy to integrate with websites, mobile applications, and other software systems.

The project successfully demonstrates how Artificial Intelligence, APIs, backend development, and databases can be combined to create a practical FAQ automation system. In the future, the

system can be enhanced with multilingual support, voice-based interaction, chat history, an admin dashboard, analytics, and more advanced AI capabilities.

Overall, the AI FAQ Assistant API provides a useful foundation for developing intelligent and automated customer-support and information-management applications.